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Units

Units

Definition. A *unit* is a standardized quantity used as a reference against which other quantities of the same kind are measured. The International System of Units (SI) defines seven base units from which all other physical quantities are derived: the metre (m) for length, the kilogram (kg) for mass, the second (s) for time, the ampere (A) for electric current, the kelvin (K) for thermodynamic temperature, the mole (mol) for amount of substance, and the candela (cd) for luminous intensity. Every physical quantity Q can be expressed as a numerical value multiplied by a unit: $Q = \{Q\} \cdot [Q]$, where $\{Q\}$ is the numerical measure and $[Q]$ denotes the unit. A change in unit produces a corresponding inverse change in numerical value, preserving the physical quantity itself.

Worked Example. Convert a speed of $v = 90 \text{ km/h}$ to SI base units (m/s). Using the conversion factors $1 \text{ km} = 10^3 \text{ m}$ and $1 \text{ h} = 3600 \text{ s}$:

$$v = 90 \frac{\text{km}}{\text{h}} \times \frac{10^3 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{90 \times 10^3}{3600} \frac{\text{m}}{\text{s}} = 25 \text{ m/s}.$$

Each conversion factor equals unity, so the physical quantity is unchanged; only its representation shifts.

Key Theorem (Dimensional Homogeneity). Any valid physical equation must be *dimensionally homogeneous*: every additive term must carry identical dimensions, and both sides of an equality must share the same dimensional form. Formally, if $[A] \neq [B]$, then the expression $A + B$ is physically meaningless. This principle, sometimes called *dimensional analysis*, provides a necessary (though not sufficient) condition for correctness and serves as a powerful consistency check. For instance, the kinematic equation $v = v_0 + at$ is verified by confirming $[\text{m/s}] = [\text{m/s}] + [\text{m/s}^2][\text{s}] = [\text{m/s}]$.

Lab Exercise. The period T of a simple pendulum is hypothesized to depend on its length L , the local gravitational acceleration g , and its mass m . Use dimensional analysis to show that T cannot depend on m , and derive that $T \propto \sqrt{L/g}$. Measure the period of a pendulum of known length and compare the predicted proportionality constant 2π against your experimental result, quantifying the percent discrepancy.